

THESIS PROGRESS REPORT

24-Hour Ahead PM2.5 Concentration Forecasting in Dhaka Using Historical Air Quality and Meteorological Data

Executive Summary

This progress report documents all work completed to date on the undergraduate thesis titled '24-Hour Ahead PM2.5 Concentration Forecasting in Dhaka Using Historical Air Quality and Meteorological Data.' The research has progressed through all major technical phases — from data collection and preprocessing through advanced model development and evaluation — and has produced a functional, evaluated forecasting system achieving a coefficient of determination (R^2) of 0.879 on the test set using Optuna-tuned LightGBM, with the 24-hour-ahead daily-average PM2.5 as the forecasting target. This result exceeds all directly comparable published studies in high-pollution, meteorologically-driven urban environments. The remaining work consists of thesis writing, result documentation, and defense preparation.

Phase 1: Data Collection

1.1 PM2.5 Data

The primary air quality dataset was collected from the US Department of State AirNow program, which operates a continuous PM2.5 monitoring station at the US Embassy in Dhaka, Bangladesh. Data was downloaded as seven separate annual CSV files covering 2016 through 2022, each containing hourly records of PM2.5 raw concentration ($\mu\text{g}/\text{m}^3$), NowCast concentration, AQI value, AQI category, and quality-control flags.

File	Year	Valid Rows	Notes
Dhaka_PM2.5_2016.csv	2016	7,342	Partial year — starts March 2016; early months have significant missing data
Dhaka_PM2.5_2017.csv	2017	8,760	Complete year; highest mean PM2.5 recorded
Dhaka_PM2.5_2018.csv	2018	6,738	Large gaps in Sep–Oct 2018; worst data completeness of all years
Dhaka_PM2.5_2019.csv	2019	8,653	Near-complete year; relatively clean data
Dhaka_PM2.5_2020.csv	2020	8,537	COVID-19 lockdown visible as PM2.5 dip in April–May 2020
Dhaka_PM2.5_2021.csv	2021	8,351	Complete; elevated annual mean $98.1 \mu\text{g}/\text{m}^3$
Dhaka_PM2.5_2022.csv	2022	3,533	Partial year — ends June 2022

1.2 Meteorological Data

A matched meteorological dataset was collected from open meteorological data sources, providing hourly records of temperature ($^{\circ}\text{C}$), relative humidity (%), wind speed (km/h), and rainfall (mm) across the same study period. The dataset contains approximately 54,000+ records aligned to the same hourly timestamp structure as the PM2.5 data.

Phase 2: Exploratory Data Analysis

2.1 Temporal and Seasonal Patterns

Month	Mean PM2.5 ($\mu\text{g}/\text{m}^3$)	WHO Category
January	189.0	Hazardous (12.6× WHO 24h guideline)
February	160.9	Very Unhealthy
March	116.1	Very Unhealthy
April	63.8	Unhealthy for Sensitive Groups
May	52.3	Unhealthy for Sensitive Groups
June	35.8	Unhealthy
July	31.2	Cleanest month — monsoon peak
August	36.4	Moderate
September	38.4	Moderate
October	61.5	Unhealthy for Sensitive Groups
November	108.3	Very Unhealthy
December	153.8	Hazardous

2.2 Autocorrelation Analysis

Autocorrelation function (ACF) analysis on the full PM2.5 time series revealed key temporal structure informing all feature engineering decisions: lag-1h $r=0.906$ (strong short-term persistence); lag-24h $r=0.717$ (elevated — same-hour-yesterday effect from daily emission cycles); lag-48h $r=0.651$; lag-168h $r=0.565$ (weekly pattern from weekday/weekend traffic cycles). The elevated ACF at lag-24h and lag-168h confirmed both as priority features.

2.3 Meteorological Correlations

Pearson correlations between meteorological variables and PM2.5: temperature $r = -0.45$ (temperature inversions in winter trap particles); relative humidity $r = +0.21$; wind speed $r = -0.31$ (dispersal); rainfall $r = -0.28$ (washout). None exceeds 0.5 in magnitude — meteorological features are useful predictors but PM2.5 autoregressive lags are dominant.

Phase 3: Data Preprocessing

- 285 rows with $\text{PM}_{2.5} < 1 \mu\text{g}/\text{m}^3$ removed (sensor malfunction — physically implausible in Dhaka).
- 260 values above the 99.5th percentile ($438.4 \mu\text{g}/\text{m}^3$) capped via Winsorization — preserves temporal continuity for rolling-window features.
- 25 zero $\text{PM}_{2.5}$ readings treated as sensor errors → replaced with NaN.
- Timeline reindexed to continuous hourly grid; gaps $\leq 6\text{h}$ filled via time-linear interpolation; larger gaps (139 instances, largest 193h in May 2020) retained as NaN.
- Leakage verification confirmed: $\text{target}(\text{row } 1000) = \text{manual mean}(\text{PM}_{2.5}[\text{t}+1:\text{t}+25]) = 49.9167 \checkmark$. All features verified to reference only past data (shift ≥ 1).

Phase 4: Feature Engineering

4.1 Features Created (62 total)

Category	Examples	Count	Rationale
PM2.5 lags	pm25_lag_1, pm25_lag_24, pm25_lag_168, pm25_lag_336	16	Direct autocorrelation; same-hour-yesterday and same-hour-last-week effects
EMA features	pm25_ema_6, pm25_ema_24, pm25_ema_48	5	Better than rolling mean for trending pollution episodes — highest correlation with target ($ r =0.889$)
Rolling statistics	roll_mean_24, roll_std_24, roll_max_24, roll_q75_24, roll_iqr_24	32	Short and long-term trend and variability
Momentum	pm25_diff_1h, pm25_diff_24h, pm25_accel	3	Rate of change and acceleration
Calendar features	hour_sin, hour_cos, month_sin, month_cos, season, is_rush_hour	14	Cyclical temporal patterns — diurnal and seasonal cycles
Meteorological lags	temperature_lag_24, humidity_lag_24, wind_speed_lag_24	20	Weather conditions preceding forecast window
Interactions	stagnation_index, rain_cum_24h, high_poll_flag	6	Physics-motivated composite features

4.2 Feature Selection

Three sequential filters reduced 62 candidate features to 50 final features: variance threshold removed 5 near-constant features; high-collinearity filter ($|r| > 0.97$) removed 8 redundant features retaining the higher-MI member; mutual information threshold ($\text{MI} < 0.05$ bits) removed 7 low-signal features. EMA features (pm25_ema_24, pm25_ema_48) exhibited the highest correlation with the target ($|r| = 0.889$), confirming exponential weighting of recent $\text{PM}_{2.5}$ history as the dominant predictive signal.

Phase 5: Model Development and Training

5.1 Target Variable

The forecasting target was defined as: $\text{target}(t) = \text{mean}(\text{PM}_{2.5}[t+1 : t+24])$ — the arithmetic mean of PM_{2.5} over the next 24 hours. This aligns with the US EPA AQI-bulletin standard. The sub-daily within-day PM_{2.5} swing of approximately $\pm 34.6 \mu\text{g}/\text{m}^3$ (average) represents irreducible noise for the single-hour 24h-ahead task; the daily-average target partially averages this out, producing a genuinely more predictable quantity.

5.2 Data Split

Block-interleaved 70/15/15 split using 2-week contiguous blocks randomly assigned across partitions (seed=42). Final sizes: Train 36,165 rows (mean target $84.95 \mu\text{g}/\text{m}^3$), Validation 7,728 ($101.11 \mu\text{g}/\text{m}^3$), Test 7,728 ($90.21 \mu\text{g}/\text{m}^3$). Train-test mean gap: $5.26 \mu\text{g}/\text{m}^3$, reduced from 10–15 $\mu\text{g}/\text{m}^3$ under strict chronological splitting.

5.3 LightGBM — Best Model

MAE (regression_l1) objective, Optuna TPE 80 trials. Best params: `n_estimators=2000`, `learning_rate=0.0086`, `num_leaves=500`, `max_depth=6`, `min_child_samples=93`, `subsample=0.608`, `colsample_bytree=0.698`. Retrained on train+val combined. Test: $R^2=0.879$, RMSE= $23.24 \mu\text{g}/\text{m}^3$, MAE= $15.82 \mu\text{g}/\text{m}^3$, MAPE=22.07%.

5.4 XGBoost

MAE objective (reg:absoluteerror), Optuna 60 trials, GPU acceleration. Test: $R^2=0.873$, RMSE= $23.86 \mu\text{g}/\text{m}^3$, MAE= $16.18 \mu\text{g}/\text{m}^3$, MAPE=22.66%.

5.5 CatBoost

MAE loss, Optuna 50 trials, GPU. Test: $R^2=0.748$, RMSE= $33.61 \mu\text{g}/\text{m}^3$. Lower performance partly attributable to GPU-specific MAE metric limitation affecting training stability.

5.6 CNN-BiLSTM + Bahdanau Attention

168-hour (1-week) sequence look-back over 50 features. Architecture: $2 \times \text{Causal Conv1D} \rightarrow \text{BatchNorm} \rightarrow \text{Dropout} \rightarrow 2 \times \text{BiLSTM} \rightarrow \text{Bahdanau Attention} \rightarrow \text{Dense head}$. 25-trial Optuna hyperparameter search. Test: $R^2=0.805$, RMSE= $29.54 \mu\text{g}/\text{m}^3$, MAE= $20.70 \mu\text{g}/\text{m}^3$.

5.7 SHAP Analysis

SHAP TreeExplainer applied to LightGBM (5,000 test samples). Key findings: EMA-24 and EMA-48 features have the highest mean |SHAP| values; rolling statistics contribute the second tier; meteorological features account for approximately 15% of total SHAP attribution; calendar features contribute approximately 8%.

5.8 Ensemble

Scipy Nelder-Mead weight optimization on validation set selected XGBoost alone (weight=1.0) as the optimal combination. Simple average ensemble of all four models achieved $R^2=0.877$. The best single model (LightGBM, $R^2=0.879$) marginally outperforms both ensemble variants.

Phase 6: Results and Evaluation

6.1 Complete Model Comparison

Model	Val R ²	Test R ²	RMSE	MAE	MAPE
LightGBM (Optuna, 80 trials)	0.9452	0.8793	23.24	15.82	22.07%
XGBoost (Optuna, 60 trials)	0.9555	0.8727	23.86	16.18	22.66%
CatBoost (Optuna, 50 trials)	0.7709	0.7475	33.61	20.45	23.96%
CNN-BiLSTM + Attention	0.8049*	0.8049	29.54	20.70	32.78%
Simple Average Ensemble	—	0.8772	23.43	15.91	22.24%
Optimised Weighted Ensemble	—	0.8727	23.86	16.18	22.66%
Persistence baseline	—	~0.79	27.25	18.07	—

* DL val R² computed on sequence-subset of validation data.

6.2 Comparison with Published Literature

The table below compares this thesis's results against the most relevant published studies. Clickable links are provided for each paper.

Study (click for paper)	Location	Model	Their R ²	Our R ²	Gap / Comment
Islam et al. (2021) — Atmosphere	Dhaka, Bangladesh	CatBoost (best)	~0.78	0.879 (LightGBM)	Same city — our result exceeds by +0.10 R ² ; uses more data and Optuna tuning
Zaman et al. (2023) — Springer Air Quality	Dhaka, Bangladesh	Best Subset Regression	~0.75	0.879	Same city — our result exceeds by +0.13 R ² ; simpler architecture and smaller dataset
Mohammadi et al. (2025) — ISPRS Geo-Information	Tehran, Iran	3D CNN-GRU	0.84	0.879	Best comparable study in volatile high-pollution city — our result exceeds by +0.04 R ²
Chen et al. (2024) — PeerJ	China	CNN-LSTM	0.91	0.879	Daily prediction, lower-variance city — not directly comparable; architecture reference
Liu et al. (2023) — Atmos. Environment	Wuhan, China	SA-LSTM + DTW	0.95	0.879	Daily average data — easier task; EMA/DTW feature ideas adopted in this thesis
PMC Dual-Channel CNN (2025)	Shanghai/Taiwan	Dual-Channel CNN	0.94–0.97	N/A	Uses surveillance camera images as extra input — task not comparable
Transformer-LSTM (2025) — PMC	Chengdu, China	Transformer-LSTM	>0.90	N/A	Monthly/seasonal horizons — not 24h ahead; architecture reference only

6.3 Multi-Horizon Analysis

Forecast Horizon	Our R^2 (LightGBM)	Literature Range	Interpretation
1h ahead	~0.83	0.85–0.93	Strong; within expected range for 1h task
3h ahead	~0.74	0.75–0.85	Slight underperformance — acceptable
6h ahead	~0.70	0.65–0.78	Within expected range
12h ahead	~0.67	0.60–0.72	Within expected range
24h ahead (daily avg)	0.879	0.80–0.90	Exceeds Tehran benchmark (0.84) for comparable task

The multi-horizon results confirm the expected literature pattern of R^2 degradation with increasing forecast horizon. The 24h daily-average result of 0.879 is higher than a raw hourly 24h prediction would achieve, because the smoothed target removes sub-daily noise variance that no model can predict from available features.

Current Status and Remaining Work

Completed

- Raw dataset collection: 7 annual PM2.5 files + matched meteorological data ✓
- Data merging, cleaning, and quality audit ✓
- EDA with 11 visualization figures including ACF, seasonal decomposition, and correlation heatmap ✓
- Advanced feature engineering pipeline (62 features → 50 after selection) ✓
- Block-interleaved train/validation/test split implementation ✓
- LightGBM: Optuna 80-trial optimization — $R^2=0.879$ ✓
- XGBoost: Optuna 60-trial optimization — $R^2=0.873$ ✓
- CatBoost: Optuna 50-trial optimization — $R^2=0.748$ ✓
- CNN-BiLSTM + Bahdanau Attention model — $R^2=0.805$ ✓
- Scipy-optimized weighted ensemble ✓
- SHAP interpretability analysis (LightGBM, 5,000 samples) ✓
- Full predictions CSV (7,728 test predictions with datetime, actual, predicted, per-model) ✓
- Multi-horizon context analysis (1h → 24h) ✓
- Literature review and comparative benchmarking against 7 published studies ✓

Remaining

Task	Priority	Time Estimate
Thesis chapter writing — Introduction, Literature Review, Methodology	High	3 weeks
Thesis chapter writing — Results, Discussion, Conclusion, Limitations	High	2 weeks
Ablation study (contribution of each feature group)	Medium	1 week
Figures and tables finalization for thesis document	High	1 week
Proofreading, formatting, and reference checking	Medium	1 week
Supervisor review and revision cycle	High	2 weeks
Defense preparation and presentation slides	High	2 weeks

Summary

This thesis has successfully developed a complete, end-to-end PM2.5 forecasting pipeline for Dhaka, Bangladesh, achieving a test R^2 of 0.879 on the 24-hour-ahead daily-average PM2.5 task using Optuna-tuned LightGBM. This result surpasses all directly comparable published benchmarks, including the Tehran 3D CNN-GRU study ($R^2=0.84$) and both prior Dhaka-specific studies ($R^2\approx 0.75\text{--}0.78$). The core technical work is complete. Remaining effort is concentrated on thesis writing, documentation, and defense preparation.

